

# **SIMATS ENGINEERING**

## **ASSESSMENT TOOL 1**

**COURSE CODE: CSA02**

**COURSE NAME: C Programming**

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### **COURSE OUTCOME COVERED**

**CO1:** *Construct structured C programs using constants, variables, data types, and preprocessor directives to address basic computational tasks. (BL2, BL3)*

*Assessment Tool Weightage: 40%*

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**QUESTIONS:**

**Total Marks:100**

Annexure A

### **PSEUDO CODE WRITING**

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#### ***Code of Conduct:***

*I \_\_\_\_\_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

*I understand that any violation of academic integrity rules will result in disciplinary action.*

***Signature of the student:***



Pseudo Code

1. Pseudocode :-

```
#define DEBUG 1
```

```
IF DEBUG == 1
```

```
    PRINT intermediate_value
```

```
END IF
```

Error:- DEBUG is 0,

2.

```
#define PI 3.14
```

```
DECLARE radius AS float
```

```
DECLARE area AS float
```

```
INPUT radius
```

```
area = PI * radius * radius
```

```
PRINT area
```

3.

```
#define RATE 1.5
```

```
DECLARE x AS int = 10
```

```
DECLARE result AS float
```

```
result = x * Rate
```

```
PRINT result
```

output : 15.0



4. # define Max 1000

DECLARE temp AS int

INPUT temp

IF temp <= MAX

Print "Valid"

ELSE

Print "Invalid"

END IF

5.

1. DEFINE Conversion rate

2. INPUT amount

3. Calculate converted value

4. PRINT result.

6.

# define MIN 10

# define MAX 50

DECLARE value AS int

INPUT value

IF (value >= MIN AND value <= MAX)

PRINT "Within range"

ELSE

PRINT "out of range"

END IF



7. # define DISCOUNT 10

# define TAX 5

# define ENABLE\_DISCOUNT 1

# define ENABLE\_TAX 1

DECLARE price AS float

DECLARE quantity AS int

DECLARE total AS float

INPUT price

INPUT quantity

total = price \* quantity

IF ENABLE\_DISCOUNT

total = total - (total \* DISCOUNT / 100)

END IF

IF ENABLE\_TAX

total = total + (total \* TAX / 100)

END IF

PRINT total

8. # define TAX 0.05

DECLARE price AS float

DECLARE total AS float

INPUT price

total = price + (price \* TAX)

PRINT total.



9. #define MODE 1

DECLARE temp AS float

DECLARE result AS float

INPUT temp

IF MODE == 1

result = (temp \* 9/5) + 32

ELSE

result = (temp - 32) \* 5/9

END IF

PRINT result

10. #define SIZE 5

DECLARE arr[SIZE]

DECLARE sum AS int = 0

DECLARE average AS float

FOR i = 0 TO SIZE - 1

INPUT arr[i]

sum = sum + arr[i]

END FOR

average = sum / SIZE

PRINT sum

PRINT average.